

Experimental Investigation and Optimization of WEDM Parameters for Titanium Alloy Considering MRR

Mohammed Aakef Farooqui¹, Ravindra Karvande², & Mohammad Irfan³

1, 2,3 Department of Mechanical Engineering, MSS's College of Engineering and Technology, Jalna, MH, India-431203

Corresponding: aakefmech226@gmail.com

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ABSTRACT

Wire Electrical Discharge Machining (WEDM) is an effective non-traditional machining process for producing complex components from difficult-to-machine materials such as titanium alloys. In the present study, an experimental investigation was carried out to optimize WEDM process parameters for machining Titanium Alloy VT20 (BT20) with the objective of maximizing material removal rate (MRR) and improving machining performance. A Taguchi L9 orthogonal array was employed to design experiments by varying pulse on-time, pulse off-time, and servo voltage at three levels. Signal-to-noise (S/N) ratio analysis, main effects plots, and analysis of variance (ANOVA) were used to identify the significance of each machining parameter. The results revealed that voltage is the most influential parameter, contributing 77.87% to the variation in MRR, followed by pulse on-time and pulse off-time. A multiple linear regression model was developed to predict MRR, and the predicted results showed good agreement with experimental values. The optimal parameter combination was found to be pulse on-time of 1.0 μ s, voltage of 35 V, and pulse off-time of 10 μ s, which produced the maximum MRR. A confirmation experiment validated the optimization with an error of only 2.23%, demonstrating the reliability of the proposed approach. The findings of this study provide useful guidelines for efficient machining of titanium alloys using WEDM.

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1. Introduction

Titanium alloys are increasingly used in aerospace, automotive, biomedical, and defense industries owing to their excellent strength-to-weight ratio, superior corrosion resistance, high fatigue strength, and ability to retain mechanical properties at elevated temperatures. Alloys such as Ti-6Al-4V and near- α titanium alloys are particularly important for structural and high-performance

applications. However, these same properties that make titanium alloys attractive for engineering applications also render them difficult to machine using conventional manufacturing processes. Conventional machining of titanium alloys is associated with several challenges, including rapid tool wear, high cutting temperatures due to low thermal conductivity, strong chemical affinity between tool and workpiece, and poor surface integrity. These limitations often result in low productivity, high machining costs, and compromised dimensional accuracy. As a result, non-traditional machining techniques have gained significant attention for effective processing of titanium alloys.

Electrical Discharge Machining (EDM) and Wire Electrical Discharge Machining (WEDM) are widely accepted non-traditional machining processes capable of machining electrically conductive materials regardless of their hardness. Material removal in these processes occurs through a series of controlled electrical discharges between an electrode and the workpiece immersed in a dielectric medium. The absence of mechanical contact eliminates cutting forces, making EDM and WEDM particularly suitable for machining titanium alloys with complex geometries, thin sections, and high precision requirements.

Wire Electrical Discharge Machining (WEDM), a specialized form of EDM, utilizes a continuously moving thin wire electrode to generate spark erosion along a programmed path. This process enables the fabrication of intricate profiles with high dimensional accuracy and minimal residual stresses. Consequently, WEDM has become an essential manufacturing technique for aerospace components, medical implants, dies, molds, and precision tooling made from titanium alloys.

Over the past decades, extensive research has been conducted to investigate the effects of EDM and WEDM process parameters on machining performance, surface integrity, and metallurgical characteristics of titanium alloys. Various modeling and optimization techniques have been employed to enhance material removal rate, improve surface quality, and reduce machining defects. Despite these efforts, the complex thermo-electrical nature of the process and strong interaction between parameters continue to pose challenges.

In this context, the present review paper aims to systematically analyze and synthesize existing research on EDM and WEDM of titanium alloys. The review focuses on machining mechanisms, material characteristics, process parameters, surface integrity, modeling and optimization approaches, industrial applications, and future research directions, providing a consolidated reference for researchers and practitioners in advanced manufacturing

2. Materials and Methods

3.1 Experimental Overview

The present study investigates the machining performance of Titanium Alloy VT20 (BT20) using the Wire Electrical Discharge Machining (WEDM) process. The objective is to identify optimal machining conditions that maximize material removal rate (MRR) and minimize surface roughness (Ra). A structured Design of Experiment (DOE) approach is adopted to systematically analyze the influence of key process parameters on machining performance.

WEDM is a non-traditional machining process in which material removal occurs through controlled electrical spark erosion between a continuously travelling wire electrode and an electrically conductive workpiece submerged in a dielectric medium. The absence of mechanical contact eliminates cutting forces, making the process suitable for machining difficult-to-machine materials such as titanium alloys.

3.2 Workpiece Material

Titanium Alloy VT20 (BT20)

Titanium alloy VT20 (BT20) is selected as the workpiece material due to its extensive application in aerospace, automotive, and biomedical components. The alloy belongs to the α - β titanium alloy category, offering a balanced combination of strength, toughness, corrosion resistance, and thermal stability.

Key characteristics of VT20 include:

- : High strength-to-weight ratio
- : Excellent corrosion resistance
- : Good high-temperature performance
- : Biocompatibility
- : Poor machinability using conventional machining methods

These characteristics justify the selection of WEDM as the machining process.

The chemical composition, mechanical properties, hardness, and thermal properties of VT20 used in this study are summarized in Tables 3.1 to 3.8, as per standard material specifications.



Figure 1 Titanium Alloy VT20 (BT20)

Brass Wire Electrode

A brass wire electrode (Cu 70% – Zn 30%) is used as the cutting tool. Brass wire is widely employed in WEDM due to its favorable electrical and thermal properties.

Key properties of the brass wire electrode:

- : High electrical conductivity
- : Good thermal conductivity
- : Low electrical resistance
- : Adequate melting point
- : Good machinability
- : Low erosion rate
- : Cost-effective and easily available

These properties enable stable spark generation and consistent machining performance.



Figure 2: Brass Wire Electrode

3.3 Experimental Machine Selection

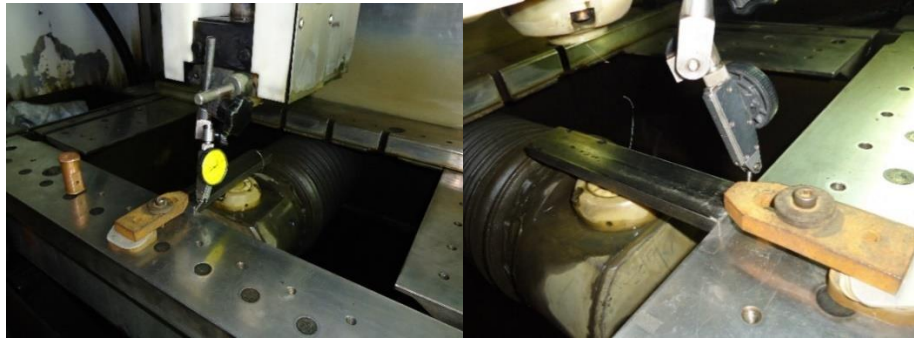


Figure 3 Figure Work piece / Part / Job & Machine Setup:

The work piece / part / job is clamped on machine table with the help of Straight Clamp & Allen screws, job zero setting is carried with the help of lever type dial indicator on upper edge right side corner for X, Y & Z direction taking rear side surface, right side surface and top surface of job as reference respectively. Machine parameters are set as per explained in Design of Experiments plan fixed parameters as it is & variable parameters i.e. Ton: Pulse on-time (μs), T off: Pulse off-time (μs), Aj: Servo Reference Voltage (v) variable at different levels.



Figure 4 WEDM machine

The experiments were performed on CharmillesRobofil 300 Series Wire Electrical Discharge Machine (WEDM) manufactured by Charmilles Technology, Switzerland which provides full

flexibility to the operator in choosing parameter values with in a wide range. A brass wire of 250 μ m diameter is used as the tool material. De-ionized water is used as the dielectric.

4. 2 Results and Discussion

4.1 Standard L9 Orthogonal Array

While there are many standard orthogonal matrices available, each of the matrices is intended for a specific number of variables and independent design levels.

The standard notation for orthogonal arrays is

$L_n(X_m)$

Where, n = Number of experiments to be carried out

X = Number of levels

m = Number of factors

The common orthogonal matrices are listed below for a quick reference.

(Arrays of 2 levels) --- L4 (23), L8 (27), L12 (211), L16 (215), L32 (231), L64 (263) etc.

(Arrays of 3 levels) --- L9 (34), L18 (21 * 37), L27 (313), L54 (21 * 325), L81 (340) etc.

(Arrangements of 4 levels) --- L16 (45), L32 (21 * 49) etc.

Note: The matrices L18 (21 * 37), L54 (21 * 325), L32 (21 * 49) etc. They are for mixed level factors.

For example, if you want to do an experiment to understand the influence of 13 different independent variables, with each variable having 3 defined values (levels), an orthogonal L27 matrix could be the right choice. The L27 OA should include the effect of 13 independent factors with 3 factor values each. This table assumes that there is no interaction between two factors, while in many cases the assumption of the interaction model is not valid and there are cases where there is clear evidence of this interaction.

In this investigation, carried out for 3 factors (pulse on time, pulse off time, velocity) each factor of 3 levels is chosen, an orthogonal L9 matrix (34) to perform the experiments. The fifth column remains free. The orthogonal L9 network was prepared using the MINITAB 17 software, specifying the input variables and their levels.

Table 1 shows an orthogonal L9 matrix. A total of 9 experiments (experiments) will be carried out and each experiment is based on the combination of the level values shown in the table. In this study, the Taguchi L9 OA is used with repetitions of the repeated reaction method to record the

data of the experimental results. Repeated measurements are taken during the same experimental tests.

The L9 OA was repeated as it is for the MRR measured during repeated runs. Each attrition index was calculated from the average of three experimental results. Each experimental test was performed six times to have two average repeated measurements.

Table 1 Standard L9 Orthogonal Array

Sr. No	Pulse on Time	Pulse off Time	Voltage
1	0.6	10	35
2	0.6	12	40
3	0.6	14	45
4	0.8	12	35
5	0.8	14	40
6	0.8	10	45
7	1.0	14	35
8	1.0	10	40
9	1.0	12	45

4.3 Model Analysis for MRR

Table 2 Cumulative data for SN Ratio MRR

Experiments	Inputs Factors			Output Responses	
Trial No.	Pulse Time on	Pulse Time off	Voltage	MRR-Ti	SNRA1
1	0.6	35	10	6.945	17.2014
2	0.6	40	12	7.945	17.9514
3	0.6	45	14	8.021	18.1718
4	0.8	35	12	7.901	17.0458
5	0.8	40	14	7.215	17.7285
6	0.8	45	10	8.912	17.9662
7	1.0	35	14	8.872	16.8046
8	1.0	40	10	9.204	17.3169
9	1.0	45	12	9.250	17.7172

S/N ratio graph of Titanium for MRR shows the with increasing energy of spark gives much material removal rate. But for SR it gives better surface finish at low range, if we give high energy it deteriorates surface finish. Because high energy removes material which form greater cavity in material which increases the Ra value.

The Table 2 and shows the L₉ orthogonal array with repeat measurement of responses for runs one to nine. Repeats of response measurement technique is used overcome the drawback of saturated design in MINITAB software. It also shows that the SN ratio for run one and ten are

same as it is calculated for the repeats measurement. The SN ratio values are calculated with help of MINITAB 17 software.

4.5 Main Effects of MRR

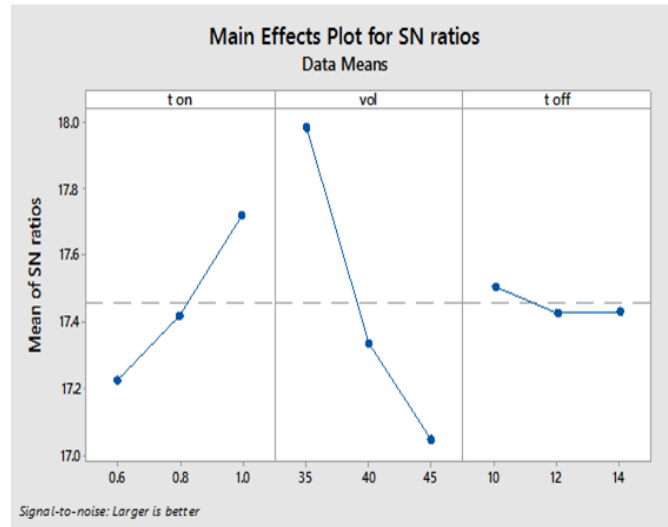


Figure 5 Main effect plots for mean of SN ratio mean of MRR

The influence of each control factor (pulse on time, pulse off time, and voltage) on material removal rate was analyzed from the S/N ratio response table, which expresses the S/N ratio at each level of control factor. The control factor influence is determined by its level difference values. A bigger control factor level difference means a greater influence on the MRR

Calculation of SN ratio:

A higher SN ratio with better characteristics was used and was calculated with the help of the MINITAB software for the experimental tests. The values of the complete SN ratio are SN ratio for the first test. From the graph, it is observed that the optimum material removal rate was in the lowest values of the in the response graph. The optimal input parameters were pulse on time 1.0 μ sec (level 1), 35 V voltage (level 2) and 10 μ sec pulse off time (level 3). The graph graphically shows the effect of the control factors on titanium material. The configuration of the process parameters with the highest ratio always provides the optimum quality with a minimum variation. The graph shows the relationship change when the control factor configuration was changed from one level to another.

From the graph of main effects for the media it is observed that the wire electro discharge machining is a function of the pulse on time, pulse off time, voltage of the material. Pulse on time is defined as the time during which the machining is performed. The machining process becomes

faster after increasing the pulse on time. By increasing the pulse on time the material removal rate increases. High pulse on time more material gets melted at the tool work piece interface, so there is not enough time for the proper flushing time but as the value of pulse off time is too short flushing to clear the debris from the inter-electrode gap between the tool and work piece, so arcing take place which result in decreasing the MR. Increase in servo voltage result in higher discharge energy per spark because of large ionization of dielectric between working gap; hence the MRR increases. But in our case The MRR decreases with increase in servo voltage this can be due to range of voltage selected, material types and strongly depend on the constant machining parameter selected for EDM machine.

These Taguchi results compared to the regression analysis show close agreement. The main effect diagrams for parameters such as pulse on time, pulse off time, voltage are obtained from the regression analysis using the Minitab Version 17 statistical software to predict the optimal level of measured response.

4.6 Analysis of Variance

The relevance of the models is tested by analysis of variance (ANOVA). It is a statistical tool for testing the null hypothesis for planned experiments, in which several different variables are studied simultaneously. ANOVA is used to quickly analyze the variances in the experiment using the Fisher test (F test). Table shows the result of the ANOVA analysis. The ANOVA analysis makes it possible to observe that the value of P is less than 0.05 in the three parametric sources. It is therefore clear that (1) the pulse on time, (2) the pulse off time, (3) the voltage of the material have an influence on the titanium material. The last column of cumulative anova shows the percentage of each factor in the total variance that indicates the degree of impact on the outcome.

Table 4.1 shows the ANOVA of titanium alloy&copper wire. The table shows that the pulse on time (21.15%), the voltage (77.87%%) and the pulse off time (6.39%) have a major influence on the material removal rate.

Table 3 ANOVA Result of MRR

Source	DF	Adj SS	Adj MS	F-Value	P-Value	% Contribution
T-on	2	0.37362	0.18681	62.81	0.016	21.15
Vol	2	1.3754	0.68877	231.21	0.004	77.87
T-off	2	0.01129	0.005646	1.90	0.04	6.39
Error	2	0.00595	0.002974			3.41

Total	8	1.76626				
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4.7 Multiple Linear Regression Models

To establish the correlation between the parameters are (1) pulse on time, (2) pulse off time, (3) voltage, of titanium material The model of multiple linear regression was obtained using the statistical software "MINITAB R 17". Terms that are statistically significant are included in the model. The final equation obtained is the following.

Mathematical models of peak voltage, pulse off time and pulse ON time is calculated using MINITAB 17 software and analysis of regression carried out to obtain the predicted value of material removal rate.

Regression Equation

$$\text{MRR} = 2.29 - 0.079 \text{ t off} + 0.0942 \text{ vol} + 3.98 \text{ t on}$$

Substituting the registered values of the variables for the previous equations, It is observed from above regression Equation that the voltage is more effective in the material removal rate of the titanium alloy. The important factor that affects material removal rate is the voltage and the coefficient associated with it is positive. This suggests that the voltage increases material removal rate.

Table 4 Comparison of Experimental and Predicted Material Removal Rate

Sr.No.	Experimental value	Predicted value
1	6.945	6.245
2	7.945	7.248
3	8.021	8.679
4	7.901	8.02
5	7.215	8.201
6	8.912	8.624
7	8.872	8.212
8	9.204	8.962
9	9.610	9.525

Surface roughness decreases for first sample and increases for ninth sample because of Experiment from 1-9 are the set of experiments from Taguchi design L9 array, every experiment has different level of parameters and having unique combination of parameter level. Also experiment no. 1 having low level of pulse on time, so surface roughness (Ra) having low value i.e better surface finish, as pulse on time having higher contribution relative to another factor.

4.8 Confirmation Test for MRR

Table 5 shows the comparison of MRR results from the mathematical model developed in the present work with values obtained experimentally respectively.

Table 5 Optimum level of parameters for Ti Alloys

Sr. No.	Parameter	Optimum level
1	Pulse on time (Level 3)	1.0
2	Voltage (Level 1)	35
4	Pulse of time (Level 1)	10

Table 6 Confirmation experiment result of Ti Alloys.

Parameter	Model value	Experimental value	Error %
MRR	9.62	9.41	2.23

5. Conclusions

In this work, Taguchi L9 orthogonal array was successfully applied to study the effect of pulse on-time (Ton), pulse off-time (Toff), and voltage on the Material Removal Rate (MRR) during WEDM of Titanium alloy (VT20/BT20). The experimental results clearly indicate that MRR increases with higher spark energy, whereas higher energy conditions tend to deteriorate surface finish, because stronger discharges create deeper craters and larger cavities on the machined surface, increasing roughness.

From the S/N ratio (larger-the-better) analysis, the optimal combination for maximizing MRR was identified as:

1. Ton = 1.0 μ s (Level 3)
2. Voltage = 35 V (Level 1)
3. Toff = 10 μ s (Level 1)

The main effects plot confirmed that Ton, Toff, and voltage significantly influence MRR, and the best performance is obtained at the parameter levels that provide stable discharge energy and effective debris flushing.

The ANOVA results revealed that voltage is the most significant parameter affecting MRR, contributing approximately 77.87% of the total variation. Pulse on-time contributed 21.15%, while pulse off-time showed comparatively lower influence (6.39%). This establishes that spark energy controlled by voltage plays a dominant role in enhancing MRR for the selected titanium alloy and machining conditions.

The regression analysis showed positive coefficients for voltage and pulse on-time, confirming that increasing these parameters increases MRR, while pulse off-time has a negative influence due to reduced discharge frequency.

Finally, the confirmation test validated the optimization and model accuracy. The predicted MRR value (9.62) closely matched the experimental value (9.41) with a low error of 2.23%, indicating that the proposed Taguchi–regression approach is reliable for predicting and optimizing MRR in WEDM of titanium alloy.

Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

References

1. Shaopeng Kang, Xianghua Zhuo, Linglei Kong, Yanxin Li and Yafeng He. "Machining characteristics and process parameter optimization of Near-dry electrical discharge milling of titanium alloy." *Scientific Reports*, 15 (2025). <https://doi.org/10.1038/s41598-025-92830-y>
2. Rajusing D. Rathod, Dinesh Kamble and N. Ambhore. "Performance evaluation of electric discharge machining of titanium alloy-a review." *Journal of Engineering and Applied Science*, 69 (2022). <https://doi.org/10.1186/s44147-022-00118-z>
3. S. Chaubey and K. Gupta. "A review on Wire-EDM of bio titanium." *Reports in Mechanical Engineering* (2023). <https://doi.org/10.31181/rme040103092023c>
4. Vivudh Gupta, B. Singh and R. Mishra. "Machining of titanium and titanium alloys by electric discharge machining process: a review." *International Journal of Machining and Machinability of Materials* (2020). <https://doi.org/10.1504/ijmmm.2020.10027192>
5. V. S. Nimbalkar and M. Shete. "Electric Discharge Machining (EDM) of Titanium Alloys: A Review." *International Journal of Engineering Research and*, 6 (2017). <https://doi.org/10.17577/ijertv6is040569>
6. Santosh Kumar and Suresh Chand Jayswal. "Machining performance analysis of wire-EDM for machining of titanium alloy using multi-objective optimization and machine learning approach." *Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering* (2024). <https://doi.org/10.1177/09544089241288009>
7. Pramanik, A. Basak and C. Prakash. "Understanding the wire electrical discharge machining of Ti6Al4V alloy." *Heliyon*, 5 (2019). <https://doi.org/10.1016/j.heliyon.2019.e01473>
8. Sarala Rubi, J. U. Prakash, S. J. Juliyana, R. Čep, Sachin Salunkhe, Karel Kouril and Sharad Ramdas Gawade. "Comprehensive review on wire electrical discharge machining: a non-traditional material removal process." *Frontiers in Mechanical Engineering* (2024). <https://doi.org/10.3389/fmech.2024.1322605>
9. Enzi, Ahmed Ghazi Abdulameer, O. Hassoon and Abdallah Hamieh. "Experimental Investigation of Manufacturing Variables Effect on Electrical Discharge Machining of Titanium Alloy." *Management Systems in Production Engineering*, 33 (2025): 202 - 211. <https://doi.org/10.2478/mspe-2025-0019>
10. M. Shehata, S. El-Hadad and Helmi Attia. "Wire Electrical Discharge Machining Process: Challenges and Future Prospects." *International Journal of Materials Technology and Innovation* (2022). <https://doi.org/10.21608/ijmti.2022.165776.1061>

11. F. Nourbakhsh, K. Rajurkar, A. Malshe and Jian Cao. "Wire electro-discharge machining of titanium alloy." *Procedia CIRP*, 5 (2013): 13-18. <https://doi.org/10.1016/j.procir.2013.01.003>
12. E. Papazoglou, P. Karmiris-Obratański, B. Leszczyńska-Madej and A. Markopoulos. "A study on Electrical Discharge Machining of Titanium Grade2 with experimental and theoretical analysis." *Scientific Reports*, 11 (2021). <https://doi.org/10.1038/s41598-021-88534-8>
13. M. Khan, P. Thejasree, M. Natarajan and K. L. Narasimhamu. "Application of a hybrid Taguchi grey approach for determining the optimal parameters on wire electrical discharge machining of Ti6Al4V." *International Journal on Interactive Design and Manufacturing (IJIDeM)*, 18 (2023): 3059 - 3076. <https://doi.org/10.1007/s12008-023-01440-3>
14. M. Thangaraj, Ramamurthy Annamalai, K. Moiduddin, M. Alkindi, S. Ramalingam and O. Alghamdi. "Enhancing the Surface Quality of Micro Titanium Alloy Specimen in WEDM Process by Adopting TGRA-Based Optimization." *Materials*, 13 (2020). <https://doi.org/10.3390/ma13061440>
15. Kamal Kumar. "Effect of Process Parameters on Surface Roughness in Wire EDM of Titanium Alloys." *INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT* (2025). <https://doi.org/10.55041/ijsrem50989>
16. Ramezan Ali Mahdavinnejad and Mohsen Asghari Ilani. "Superior advance research in the electro-discharge machining of Ti alloys: Review." *International Journal of Scientific Research in Mechanical and Materials Engineering* (2019). <https://doi.org/10.32628/ijrmme19339>
17. Pramanik, A. Basak, G. Littlefair, S. Debnath, C. Prakash, Meinam Annebushan Singh, D. Marla and R. Singh. "Methods and variables in Electrical discharge machining of titanium alloy – A review." *Heliyon*, 6 (2020). <https://doi.org/10.1016/j.heliyon.2020.e05554>
18. Jaber E. Abu Qudeiri, A. Mourad, Aiman Ziout, M. H. Abidi and A. Elkaseer. "Electric discharge machining of titanium and its alloys: review." *The International Journal of Advanced Manufacturing Technology*, 96 (2018): 1319 - 1339. <https://doi.org/10.1007/s00170-018-1574-0>
19. M. Mahalingam. "Review on Wire Electric Discharge Machining Process of Titanium Alloy." *International Journal of Scientific Research in Science and Technology*, 4 (2018): 1848-1854. <https://doi.org/10.32628/ijrst184975>